

21/22/HND/20/023

FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF ELECTRICAL/ELECTRONICS ENGINEERING
TECHNOLOGY

2ND /4TH SEMESTER EXAMINATION 2021/2022 SESSION
DECEMBER, 2022

PROGRAMME: HND II
COURSE TITLE: DIGITAL COMMUNICATION III
COURSE CODE: EEE 445
TIME ALLOWED: 2hrs 30mins
INSTRUCTION:

Questions paper has two sections:

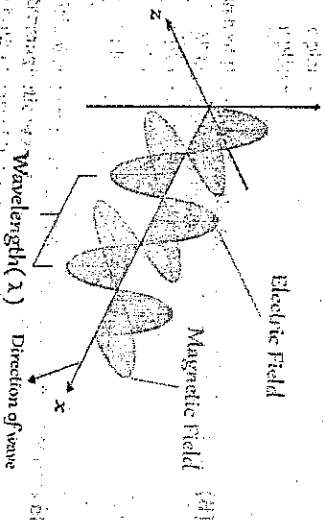
Section A: OBJECTIVES & Section B: Essay

Answer all questions in Section A, with any three (3) in section B

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SECTION A: OBJECTIVES (1 mark each)

Use the EM wave signal information below to answer Q.1-4



Q2. Wave direction of propagation is...

- A. All listed planes below
B. z direction
C. y direction
D. x direction

Q3. Which of the plane lies the Electric Field Component (E)

- A. All listed planes above
B. z plane
C. x plane
D. y plane

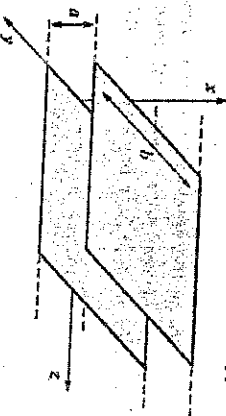
Q4. Which of the plane lies the Magnetic Field Component (H)

- A. All listed planes above
B. y plane
C. x plane
D. z plane

Q5. Which of the following is NOT true about the properties of an electromagnetic wave?

- A. Have both electric (E) & magnetic (M) components.
B. Affected by the medium through which it travels
C. Travel in free space or guided path (Waveguide)
D. Increases as the distance increases

Use the diagram below to answer Questions 6 - 7.



Q6. In the parallel-plate waveguide, the plates are assumed to be infinite in extent in...

- A. x and z directions.
B. x and y directions.
C. y and z directions.
D. All listed planes above

Q7. The plates are finite in in which the waves are guided.

- A. z direction
B. y direction
C. x direction
D. All listed planes above

Q8. For perfectly conducting plates with source-free, simple, and lossless assumption. Which of the following applies?

- A. ϵ and μ non-constants and $\alpha = 0$, μ and ϵ complex

- B. ϵ and μ are complex and $\alpha = 0$, μ and ϵ are real
C. ϵ and μ are constants and $\alpha = 0$, μ and ϵ are real
D. None of the above

Answer Q.9 - 12 using field solutions for TE, TM & TEM modes.

Q9. Transverse Electric Mode: which of the following is true?

- A. $E_z \neq 0$, $H_z = 0$
B. $E_z \neq 0$, $H_z \neq 0$
C. $E_z = 0$, $H_z \neq 0$
D. $E_z = 0$, $H_z = 0$

Q10. Transverse Magnetic Mode: which of the following is true?

- A. $E_z = 0$, $H_z = 0$
B. $E_z = 0$, $H_z \neq 0$
C. $E_z \neq 0$, $H_z = 0$
D. $E_z \neq 0$, $H_z \neq 0$

Q11. Transverse Electromagnetic (TEM) Mode: which of the following is true?

- A. $E_z \neq 0$, $H_z = 0$
B. $E_z = 0$, $H_z = 0$
C. $E_z = 0$, $H_z \neq 0$
D. $E_z \neq 0$, $H_z \neq 0$

Q12. In waveguides, the equations are subject to which of the following boundary conditions at the conductor surfaces?

- A. $E_{tangential} \neq 0$, $H_{normal} \neq 0$
B. $E_{tangential} = 0$, $H_{normal} = 0$
C. $E_{tangential} \neq 0$, $H_{normal} \neq 0$
D. $E_{tangential} \neq 0$, $H_{normal} = 0$

Q13. Which of the following is NOT the type of waveguide?

- A. Rectangular waveguide
B. Triangular-shape waveguide
C. Circular waveguide
D. Double ridged

Answer Q.14 - 17 with TE modes

$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$
$n=1$ 3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013
$n=2$ 7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349
$n=3$ 10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682

Q14. Which of this values of the zeroes of the dominant mode?

- A. 15.2682
B. 1.8412
C. 3.0542
D. 3.8318

Q15. Which of these following corresponds to the dominant mode designation?

- A. TE_{01}
B. TE_{11}
C. TE_{21}
D. TE_{01}

Q16. Which of the following value is the second dominant mode of the table?

- A. 3.0542
B. 15.2682
C. 1.8412
D. 3.8318

Q17. Which of the following corresponds to the second dominant mode designation?

- A. TE_{21}
- B. TE_{03}
- C. TE_{11}
- D. TE_{01}

Answer Q 18 – 21 with TM modes

χ_{nm}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038

Q18. Which of these values of the zeroes is the dominant mode?

- A. 2.4049
- B. 17.0038
- C. 3.8318
- D. 5.1357

Q19. Which of the following corresponds to the dominant mode designation?

- A. TM_{01}
- B. TE_{01}
- C. TM_{11}
- D. TM_{21}

Q20. Which of these values is the second dominant mode of the table?

- A. 3.8318
- B. 17.0038
- C. 2.4049
- D. 5.1357

Q21. Which of the following corresponds to the second dominant mode designation?

- A. TM_{21}
- B. TE_{03}
- C. TM_{01}
- D. TM_{11}

Q22. Which of the following is NOT a waveguide component?

- A. Attenuators, Phase shifters
- B. Waveguide bends
- C. Match load
- D. Diode

Q23. Which of the following is NOT a Microwave tube?

- A. Helix TWT
- B. Two-cavity Klystron
- C. Magnetron
- D. Thermistor

Q24. Which of the following are Microwave Semiconductor device?

- A. Dominant mode
- B. Attenuators
- C. Phase shifters
- D. Gunn Diode

Q25. Which of the following is a function of a RADAR system?

- A. Gives the range of the target from the Radar station
- B. Gives the radial velocity of the target
- C. Detects the presence of the target.

D. All listed options above

Q26. In the RADAR system, the term pulse repetition time (abbreviated PRT) is mathematically given as time:

- A. For the pulse to reach the receiving antenna
- B. For pulse transmission to hit the target and return
- C. From the beginning of the first pulse to the beginning of the next
- D. All the above

Q27. The range of the RADAR system can be measured in which of the following unit?

- A. Nautica Miles (nmi)
- B. Kilometres (km)
- C. All the listed options
- D. Metres (m)

Q28. Which of the following body are tasked with the allocation and regulating of the frequency bands for the Radar system?

- A. Radio Broadcasting
- B. Military Defence Services
- C. International Telecommunication Union (ITU)
- D. Airport Authority

Q29. Which of the following is NOT an application of RADAR systems?

- A. Air Traffic Control
- B. Military Defence
- C. Radio Broadcasting Services
- D. Remote Sensing

Q30. Law Enforcement & Highway Safety can use the RADAR system to do which of the following?

- A. For warning of a pending collision
- B. Actuating airbag or warning of obstruction
- C. To measure the speed for enforcing speed limits
- D. Demobilised the moving target

Q31. RADAR system can be used in Air Traffic Control to do of the following?

- A. Demobilised the moving target approaching
- B. To safely control air traffic in the airports/enroute
- C. Broadcast to the passenger's flight information
- D. All of the above-listed options

Q32. Which of the following best describes what a fibre optics is...?

- A. It is made of low-loss materials such as silica or glass.
- B. All of the listed A, C & D options.
- C. It has a central core in which the light is guided.
- D. Cylindrical dielectric waveguide.

Q33. Fibre optics core and cladding have different refractive indices... which of the following is TRUE about the core and cladding of a fibre optics...?

- A. Cladding also have a refractive index, n_2
- B. All of the listed options below
- C. refractive index n_1 slightly higher than n_2
- D. Core has a different refractive index, as n_1

Q34. Which of the following can be used to discuss the geometry of optical fibre?

- A. Maxwell's law.

- Q34. Which of the following is NOT a type of satellite?
- A. LEO
 - B. MEO
 - C. GEO
 - D. ITO

- Q35. For optical fibre structure and physics, because $n_1 > n_2$ then it is always that
- A. $f_1 \neq f_2$
 - B. $f_1 > f_2$
 - C. $f_1 < f_2$
 - D. $f_1 = f_2$

- Q36. Which of the following is NOT a satellite orbit?
- A. LEO
 - B. MEO
 - C. NEO
 - D. GEO

- Q37. Which of the following defined the distance of LEO satellites from the earth's surface?
- A. 37,86 km.
 - B. 10,000 km.
 - C. 500-1500 km.
 - D. None of the above.

- Q38. As frequency allocation for satellite services by ITU requires international coordination and planning, the whole world is divided into how many regions?
- A. One region
 - B. Two regions
 - C. Three regions
 - D. Four regions

- Q39.
- A.
 - B.
 - C.
 - D.

- Q39. Which of the following is NOT the applications of satellite system?
- A. Global Mobile Communications
 - B. Radio and TV broadcast
 - C. Highway/Law Enforcement
 - D. Navigation services.

- Q40. Which of the following frequency band is NOT allocated for satellite services?
- A. L-band 1.0 – 2.0GHz.
 - B. C-band 4.0 – 8.0GHz.
 - C. mm-Wave Frequencies
 - D. Ku-band 12 – 18GHz.

SECTION B: ESSAY (20 marks for each question)
Instruction: Answer any three (3) questions from this section.

Q.1 (a) What is electromagnetic wave?
 (b) List any four (4) properties of electromagnetic waves and define any two (2).

(5 Marks)

Q.2 (c) Show mathematically, the relationships among the listed properties above. (10 Marks)
 (a) What is waveguide? (5 Marks)

(b) Using the values of zeros as presented in the Bessel function and Bessel function derivative tables below, copy and complete the table. (5 Marks)

TE modes

X_{mn}'	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682

TM modes

X_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038

Note: From the tables above, the 1st dominant mode is the TE_{11} mode, followed in order by the TM_{01} mode, then the TE_{21} mode, and the TE_{01} mode. Hence the complete table is given below:

S/No.	Order	Mode	Designation	Values of Zeros
1.	1 st Dominant		TE_{11}	
2.		TM		2.4049
3.	3 rd Dominant		TE_{21}	
4.		TE		3.8318

Q.3 (c) Draw a complete diagram of two cavity klystrons, list five components. (10 Marks)
 (a) What is a RADAR system? (5marks)

(b) Draw a block diagram of a RADAR system and explain any two parts. (5marks)

(c) List five (5) FIELD applications (with 3 sub-applications) of a RADAR system. (10marks)

Q.4 (a) Derive the RADAR equation (range performance) and define the parameters. (10marks)

(b) Calculate the maximum range of a radar that operates at a frequency of 20GHz with, a peak pulse power of 1.2MWatts. If the antenna effective area is $10m^2$ and the area of the target is $40m^2$ and the minimum receivable power of 0.2n watts. (10marks)

Q.5 (a) Briefly write a short note about the satellite system. (5 Marks)

(b) Briefly explain geostationary orbit? (5 Marks)

(c) List any four (4) applications of a satellite system and briefly explain any 2. (10 Marks)

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 2ND & 4TH SEMESTER EXAMINATION, 2020/2021 ACADEMIC SESSION

FEBRUARY 2022

PROGRAM: HND II Electrical Engineering Technology
COURSE TITLE: Digital Communication III
COURSE CODE: EEE 445

UNIT: 3

TIME ALLOWED: 3 Hours

INSTRUCTION: Attempt any Five (5) Questions only in both sections (A & B).

SECTION A: OBJECTIVES (2 marks each)
QUESTION 1

1. The following is **NOT** true about the properties of an electromagnetic wave?
 - A. ✓ Electromagnetic wave power increases as the distance increases.
 - B. Travel in free-space or guided path (waveguide).
 - C. ✓ Affected by the properties of the medium through which it travels.
 - D. Have both electric field (E) & magnetic field (M) components.

2. In an ideal parallel-plate waveguide, the plates are assumed to be infinite in extent in the
 - A. ✓ y and z directions.
 - B. x and y directions.
 - C. x and z directions.
 - D. all directions.

Use the diagram in Fig. 1 to answer Question 3:

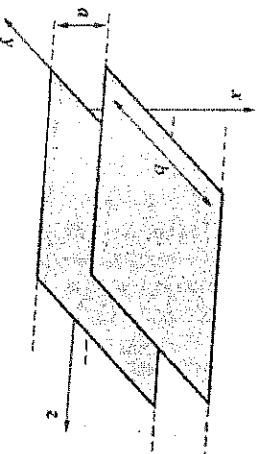


Fig. 1

3. As figure above shown, the plates have an infinite extent in the in which the waves are to be guided?
 - A. ✓ z directions
 - B. y directions
 - C. x directions
 - D. r directions

4. For field solutions of an electromagnetic wave, which of the following is true about Transverse Electric (TE) Mode:

A. $E_z = 0,$	$H_z \neq 0$
B. $E_z \neq 0,$	$H_z = 0$
C. $E_z \neq 0,$	$H_z = 0$
D. ✓ $E_z \neq 0,$	$H_z \neq 0$

5. For field solutions of an electromagnetic wave, which of the following is true about Transverse Electromagnetic (TEM) Mode:

A. ✓ $E_z = 0,$	$H_z = 0$
B. $E_z \neq 0,$	$H_z = 0$
C. $E_z = 0,$	$H_z \neq 0$
D. $E_z \neq 0,$	$H_z \neq 0$

6. For a metallic waveguide, the solutions of the wave equations are subject to which of the following boundary conditions at the conducting surfaces?

A. $E_{\text{tangential}} \neq 0$	$H_{\text{normal}} \neq 0$
B. ✓ $E_{\text{tangential}} = 0$	$H_{\text{normal}} = 0$
C. $E_{\text{tangential}} = 0$	$H_{\text{normal}} \neq 0$
D. $E_{\text{tangential}} \neq 0$	$H_{\text{normal}} = 0$

7. Which of the following is NOT type of waveguide?

- A. Rectangular waveguide
- B. Diamond-shape waveguide
- C. Circular waveguide
- D. Double-ridged waveguide

Use the following Circular waveguide Bessel function table to answer question 8 & 9

χ_{mn}'	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013	...
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349	...
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682	...

8. From the Bessel function of the circular waveguide TE modes above, which of the following value of the zeroes?

- A. 15.2682
- B. 1.8412
- C. 3.0542
- D. 3.8318

9. Which of the following correspond to the dominant mode designation?

- A. TE_{03}
- B. TE_{11}
- C. TE_{21}
- D. TE_{01}

QUESTION 2

Use the following Circular waveguide Bessel function table to answer question 10 & 11

χ_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	...
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893	...
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	...

10. From the Bessel function of the circular waveguide TM modes above, which of the following value of the zeroes of the dominant mode?

- A. 17.0038
- B. 2.4049
- C. 3.8318
- D. 5.1357

11. Which of the following correspond to the dominant mode designation?

- A. TE_{03}
- B. TM_{11}
- C. TM_{01}
- D. TM_{21}

12. Which of the following is NOT a waveguide component?

- A. Match load
- B. Transducer
- C. Waveguide bends
- D. Attenuators, Phase shifters

13. Which of the following is NOT a Microwave tubes?

- A. two cavity Klystron
- B. Schottky Diode

- C. Magnetron
- D. Helix TWT

14. Which of the following is NOT a Microwave Semiconductor device?

- A. Gunn Diode
- B. Colour Coded Resistor
- C. Schottky Diodes
- D. PN Junction Diode

15. RADAR can be use in the Air Traffic Control to do which the following?

- A. Demobilised the moving target
- B. Actuating air bag or warning of obstruction
- C. To safely control air traffic in the vicinity of the airports and enrooted
- D. Broadcast to the passengers about their flight's information

Handwritten: Satellite mode with the earth surface at the speed — Geo

16. Which of the following is NOT an orbit of a satellite system?

- A. Geosynchronous Earth Orbit (GEO)
- B. Low Earth Orbit (LEO)
- C. Medium Earth Orbit (MEO)
- D. Center Earth Orbit (CEO)

Handwritten: at least a fiber optic

17. RADAR can be used by Law Enforcement & Highway Safety to do the following, EXCEPT?

- A. To measure the speed for enforcing speed limits
- B. Actuating air bag or warning of obstruction
- C. For warning of pending collision
- D. Demobilised the moving target

18. Fibre optics core and cladding have different refractive indices... which of the following is TRUE about the core and cladding of a fibre optics...?

- A. Core have different refractive index, as n_1 .
- B. Cladding also have a refractive index, n_2 .
- C. refractive index n_1 slightly higher than n_2 .
- D. All of the above listed options

19. Which of the following can be used to discuss the geometry of optical fiber?

- A. Biot Savart law.
- B. Maxwell's law.
- C. Faraday's law.
- D. Snell's law

20. Which of the following is NOT a Components of two cavity klystron...?

- A. Collector
- B. Buncher Cavity.
- C. Catcher cavity.
- D. None of the above

SECTION B: ESSAY Consisting of 6 questions (20 marks each)

Instruction: Answer any two (2) QUESTION from this section.

Q.3 (a) Briefly explain a waveguide? (5marks)

(b) List any four waveguide and briefly explain any two (2) (10marks)

(c) Design an air-filled circular waveguide yielding a frequency separation of 1GHz between the cut-off frequencies of the dominant mode and the next highest

mode. [Use the Bessel function table below]

TE modes

(5marks)

χ'_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013	...
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349	...
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682	...
$n=4$	13.3234	11.7919	13.3234	14.8071	16.3001	17.7793	19.2446	...
$n=5$	16.4706	15.0701	16.4706	18.0803	19.6248	21.1065	22.5271	...
$n=6$	19.6158	18.1541	19.6158	21.2539	22.9815	24.6525	26.2661	...

TM modes

χ_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	...
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893	...
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	...
$n=4$	11.7919	13.3234	14.8071	16.2426	17.6281	19.0646	20.4521	...
$n=5$	14.9331	16.4706	18.0803	19.6248	21.1065	22.5271	23.9086	...
$n=6$	18.0803	19.6158	21.2539	22.9815	24.6525	26.2661	27.8298	...

Note that the dominant mode in a circular waveguide is the TE_{11} mode, followed in order by the TM_{01} mode, the TE_{21} mode and the TE_{01} mode.

Q.4

- Briefly explain what is an Optical fiber? (5marks)
- List any four components of waveguide and briefly explain any two (2) (5marks)
- Draw a complete diagram of two cavity klystron, list the characteristics, and its five components. (10marks)

Q.5

- Define a RADAR system? ✓ (10marks)
- Draw a block diagram of a RADAR system and explain any two parts. (5marks)
- List five (5) FIELD applications (with 3 sub-applications) of a RADAR system. (5marks)

Q.6

- Write the simple form of RADAR equation (range performance) and define the parameters. (10marks)
- Calculate the maximum range of a radar which operates at a frequency of 20GHz, peak pulse power of 1.2MWatts. If the antenna effective area is $10m^2$ and the area of the target is $40m^2$ and the minimum receivable power of 0.2n watts. (10marks)

Q.7

- Briefly write a short note about a satellite system? (10marks)
- Explain in detail, what is geostationary orbit? (5marks)
- List any four (4) applications of a satellite system and briefly explain any 2. (10marks)



THE FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF COMPUTER SCIENCE
END OF 2nd AND 4th SEMESTER EXAMS 2021/ 2022 SESSION TIME: 2:30HRS

55/22/Answer/20/623

Programme: HND II ELECTRICAL ELECTRONICS.
Course Code: ICT 301.

Course Title: INTRODUCTION TO C++ PROGRAMMING.

Instructions: Answer any 5 questions. All Question Carry Equal Marks (12mks each).

QUESTION ONE (12mks each)

- Explain the term C++.
- Why C++ is called an OOP?
- Write short note on any six (6) features of C++.

4mks
2mks
6mks

QUESTION TWO

- Why C++?
- List & explain any three (3) types of Data Types you know.
- Using a simple program, show why is C++ called partially OOP.

2mks
6mks
4mks

QUESTION THREE

- Show syntax for Variable Declaration.
- What are the significances of function in C++ program?
- Differentiate between Local & Global Variable Declaration with example.

4mks
3mks
5mks

QUESTION FOUR

- What is an operator in C++?
- Differentiate between Data Types & Variable.
- Write short note on any four (4) types of operators in C++.

1mks
3mks
8mks

QUESTION FIVE

- Define the term function as apply to C++.
- Distinguish between keywords & identifies with examples.
- Write short note on the following: (i) Token (ii) Constant (iii) Escape sequence.

1mks
5mks
6mks

QUESTION SIX

- Distinguish between Passing & Default Argument.
- Write short note on (i) Classes (ii) Array (iii) Pointer.
- Using a program, demonstrate the working of Input & Output Operator.

2mks
6mks
4mks

QUESTION SEVEN

- How does pointer works in C++.
- Differentiate between One & Multi-dimensional arrays with example.
- In a tabular form differentiate between Library & User-defined Functions.

3mks
4mks
5mks

EST/EE/4203/20/623

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY

END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
DECEMBER 2022

PROGRAM: HNDII Electrical Engineering Technology
COURSE TITLE: Computer Hardware Maintenance and Repair

UNIT: 2

COURSE CODE: EEE 447

TIME ALLOWED: 2 Hours

INSTRUCTION: Attempt any Four (4) Questions only. All question carries equal marks, do not write on the question paper.

QUESTION ONE

- a) Define the term "Personal Computer" (2marks)
- b) Give and briefly explain the three (3) Types of Computers (8marks)
 - i. Define the term "Computer System Bus". (3marks)
 - ii. Briefly explain the three (3) types of CPU buses. (3marks)
- d) Define the following: i) Peripheral ii) Accessory iii) Byte iv) Character (6marks)

QUESTION TWO

- a) Define the following terms:
 - i. Cylinder
 - ii. Cluster
 - iii. Sector
 - iv. Storage space
 - v. Track
- b) State the four functional sections of a CPU. (5marks)
- c) Briefly explain the three (3) types of CPU buses. (8marks)
- d) Give the full meaning of the following acronyms with respect to Computer motherboard architecture; (3marks)
 - i) BIOS ii) AGP iii) IDE iv) USB v) VGA vi) PCI vii) DIMM viii) SIMM ix) PCMCIA x) CMOS

QUESTION THREE

- a) Define the term "Computer Software" and state the different types of software. (5marks)
- b) State four (4) examples each of the different types of software as stated in question 3(a) above. (12marks)
- c) State the three (3) basic characteristics of a hard drive for optimal performance. (3marks)
- d) Name any five (5) supplementing tools used in troubleshooting computer system. (5marks)

QUESTION FOUR

- a) State any six (6) basic troubleshooting tools. (6marks)
- b) Briefly describe the procedure for troubleshooting a dead computer system. (7marks)
- c) The symptoms given in (i) and (ii) below were observed on a faulty computer system, in a tabular form write out the checks and actions to be performed for each symptom in order to repair this computer system.

- (i) Power supply appears dead. Screen appears dead. Fan not running (6marks)
- (ii) System power up, screen alive BIOS does not register drive. (6marks)

QUESTION FIVE

- a) Define Electrostatic Discharge (ESD) and state six (6) ways of reducing ESD (7marks)
- b) Briefly explain the causes of electrostatic discharge. (6marks)
- c) State any six (6) importance of preventive maintenance. (6marks)
- d) Briefly describe the procedure for preventive maintenance. (6marks)

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THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 2ND & 4TH SEMESTER EXAMINATION, 2020/2021 ACADEMIC SESSION

JANUARY 2022

PROGRAM: HND/II Electrical Engineering Technology.
COURSE TITLE: Computer Hardware Maintenance and Repair.
COURSE CODE: EEE 447
TIME ALLOWED: 2 1/2 Hours

UNIT: 1

INSTRUCTION: Attempt any Four (4) Questions only.

All question carries equal marks. Do not write on the question paper

QUESTION ONE

- a) Write short notes on the following: -
- i. Storage device (2marks)
 - ii. Input/output device (2marks)
- b) Give two (2) examples each for the following:
- i. Storage device (2marks)
 - ii. Input/output device (2marks)
- c) State the four functional sections of a CPU. (4marks)
- d) Define the term “Computer System Bus”. (8marks)
- e) Briefly explain the three (3) types of CPU buses. (4marks)
- (3marks)

QUESTION TWO

- a) Define the following:
- (i) Peripheral. (1 1/2 marks)
 - (ii) Accessory. (1marks)
 - (iii) Byte. (1 1/2 marks)
 - (iv) Character. (1marks)
- b) Write the full meaning of the following acronyms with respect to Computer motherboard architecture;
- (i) CPU (1mark)
 - (ii) SIMM (1mark)
 - (iii) BIOS (1mark)
 - (iv) PCI (1mark)
 - (v) EISA (1mark)
 - (vi) USB (1mark)
 - (vii) IDE (1mark)
 - (viii) AGP (1mark)
 - (ix) ROM (1mark)
 - (x) CMOS (1mark)
- c) Define the following terms:
- i. Cylinder (2marks)
 - ii. Cluster (1marks)
 - iii. Sector (1marks)
 - iv. Storage space (2marks)
 - v. Track (1marks)
- d) State the three (3) basic characteristics of a hard drive for optimal performance. (3marks)

QUESTION THREE

- a) Define the term “Computer Software”. (5marks)
- b) state the different types of software (5marks)
- c) State four (4) examples each of the different types of software as stated in question 3(a) above (12marks)
- d) Explain the term “Computer Operating System” (3marks)

QUESTION FOUR

- a) State any six (6) basic troubleshooting tools. (6marks)
- b) Briefly describe the term "**Interrupt Request Line**". (6marks)
- c) State the first four (4) Interrupt Request Line lines according to priority (5marks)
- d) State five (5) steps to check when hardware problem is discovered after booting. (8marks)
- e) Briefly describe the procedure for troubleshooting a dead computer system

QUESTION FIVE

- a) Define Electrostatic Discharge (ESD). (1 mark)
- b) Briefly explain the causes of electrostatic discharge. (6marks)
- c) state six (6) ways of reducing Electrostatic Discharge ESD (6marks)
- d) The symptoms given in (i) and (ii) below were observe on a faulty computer system, in a tabular form write out the checks and actions to be performed for each symptom in order to repair this computer system. (6marks)
- (i) Power supply appears dead. Screen appears dead. Fan not running
- (ii) System power up, screen alive BIOS does not register drive. (6marks)

QUESTION SIX

- a) Briefly explain the main functions of the BIOS (8marks)
- b) State six (6) checks performed for problems that occur during the POST (6marks)
- c) State how to access/run the CMOS setup program. (3marks)
- d) Describe any six (6) Setup menu items found on a typical CMOS environment. (6marks)
- e) State the two major types of security password associated with the BIOS. (2marks)

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THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY

END OF 2ND & 4TH SEMESTER EXAMINATION, 2020/2021 ACADEMIC SESSION

FEBRUARY 2022

PROGRAM: *HNDII Electrical Engineering Technology*
COURSE TITLE: *Electronic Maintenance & Repairs.*
COURSE CODE: *EEE 446*
TIME ALLOWED: *2 1/2 Hours*

UNIT: 1

INSTRUCTION: *Attempt any Four (4) Questions only*

All question carries equal marks. Do not write on the question paper

- Question 1 ✓
- (a) Explain in brief the testing methods necessary to locate the following faults in electronic circuits
 - (i) short circuit
 - (ii) Open circuit
 - (b) Describe briefly how to use a digital multimeter to identify the type/pin configurations of a BJT transistor.

Question 2 ✓ (15 mks)

- (a) (i) Describe in brief how to use a signal generator and oscilloscope only to troubleshoot the two-stage transistor amplifier shown in figure Q.2a. (3 mks)

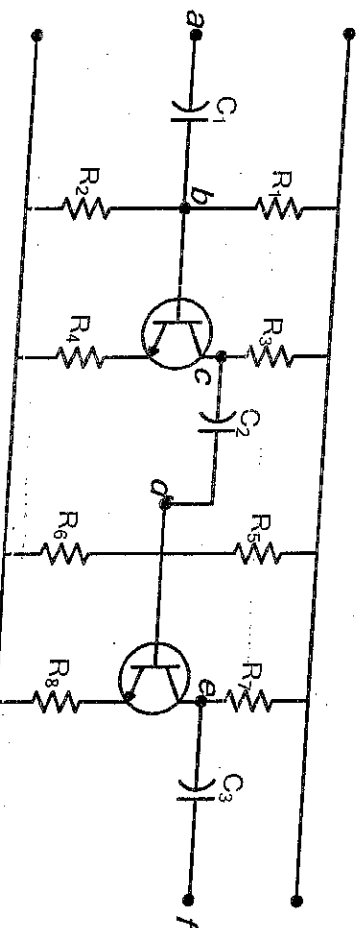


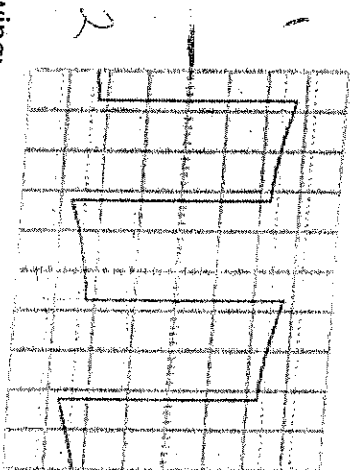
FIGURE Q-2a

- (ii) In Fig.Q2a above, a pure sine wave of amplitude 1.0 mV is injected at point A. Use a diagram to show how the systems output waveform can be observed. (8mks)
- (iii) Comment on the systems waveform quality and amplitude at points a, b, c, d, e, and f when, system is okay (6mks)

- (b) State the safety precautions to be taken during testing of electronic equipment. (8 mks)

Question 3

- (a) Fig.Q.3a shows the waveform displayed by an oscilloscope with a vertical setting of 4.0 V per division and a time base setting of 0.75 millisecond per division.



Determine the following:

- frequency (f)
- V_{peak}
- V_{P-P}
- V_{rms}

$$f = \frac{1}{T}$$

$$V_{peak} = 2.75 \times 4 = 11$$

$$V_P = 11$$

$$V_{PP} = 2 \times V_P$$

$$V_{rms} = \frac{V_P}{\sqrt{2}}$$

- (7marks)
- (3marks)
- (2marks)
- (3marks)

(b) Sketch and clearly label the various stages in the block diagram of a radio receiver (10marks)

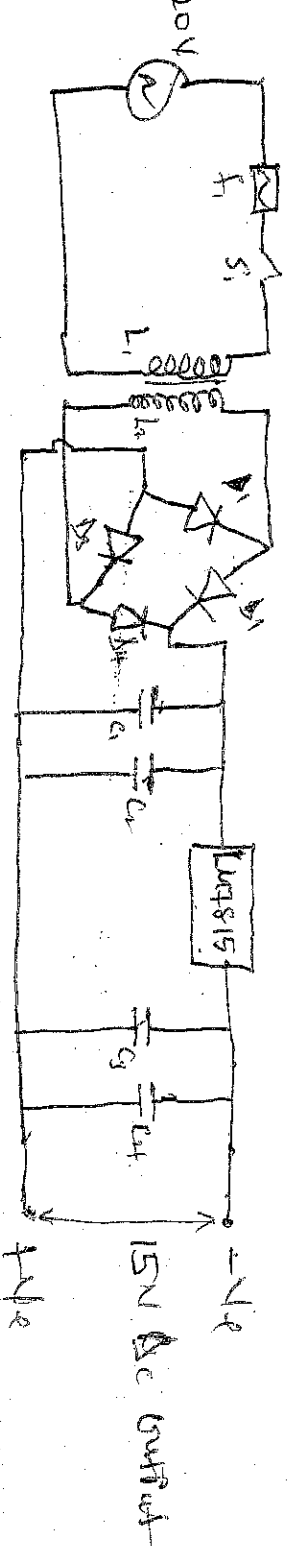
- What is alignment? (5marks)
- List the steps for typical CD player alignment. (8marks)
- Describe how to align a satellite receiver dish. State clearly why trees should be avoided. (12marks)

Question 5 ✓

- Draw a well labeled 15.0 V dc regulated power supply that use the following components – (10marks)
 - 1) Switch
 - 2) Fuse
 - 3) transformer
 - 4) rectifying diodes
 - 5) Smoothing capacitor
 - 6) 15 V regulator
- If no output from the power supply, describe briefly the steps you would take to detect the fault. (15 marks)

Question 6

- List the basic components of a typical public address system (3 mks)
 - Describe how best to troubleshoot the system if there is no audio output. (12 mks)
- Sketch and clearly label the various stages in the block diagram of TV receiver. (10 mks)



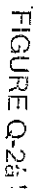
UNIT: 1

Questions only. All question carries equal marks. Do not write on the question paper

- (5 mks)
(5 mks)

(15 ml's)

- (3 mī's)



(14 mks)
(8 mks)

1

- per division



Determine the following:

- (i) frequency (f)
- (ii) V_{peak}
- (iii) $V_{\text{P-P}}$
- (iv) V_{rms}

- (b) Sketch and clearly label the various stages in the block diagram of a radio receiver

(6mks)
(3mks)
(3mks)
(3 mks)
(10 mks)

QUESTION 4

- (a) Differentiate between Alignment and calibration as regards Electro mechanical systems.
- (b) Describe how to align a satellite receiver dish. State clearly why trees should be avoided.

(10 mks)
(12mks)

QUESTION 5

- (a) (i) Draw a well labeled 12.0 V dc regulated power supply that use the following components –
 - (a) Switch
 - (b) Fuse
 - (c) transformer
 - (d) rectifying diodes
 - (e) Smoothing capacitor
 - (f) 9 V regulator
- (b) If no output from the power supply, describe briefly the steps you would take to detect the fault.

(10 mks)
(15 mks)

QUESTION 6

- (a) (i) List and define the basic components of a typical public address system
- (ii) Describe how best to troubleshoot the system if there is no audio output.
- (b) Sketch and clearly label the various stages in the block diagram of TV receiver.

(3 mks)
(12 mks)
(10 mks)

The Federal Polytechnic, Mubi
Department of Mathematics and Statistics,
2nd And 4th Semester Examinations, 2020/2021 Session
Departments: EE/ME/AB/ET Course Code: MTH 313/413/412
Course Title: Statistical Methods
Instruction: Answer any Four (4) questions. Time allowed: 3hrs.

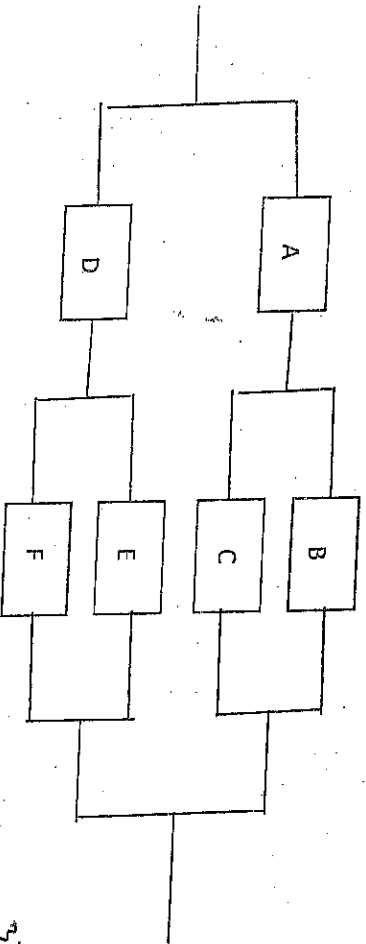
1 (a) What is the use of quality control?

(b) Quality control is to be applied to the production of tablets by controlling the weights of the tablets. Ten samples each containing four tablets were taken at random with weights as follows:

Sample No	1	2	3	4	5	6	7	8	9	10
	3.7	3.9	3.0	3.4	3.6	3.5	3.2	2.9	3.0	3.8
	2.9	3.0	3.7	3.2	3.1	3.3	3.2	3.0	3.1	3.7
	3.2	3.3	3.8	3.4	3.1	3.2	3.3	3.2	3.3	3.2
	3.3	3.1	3.1	2.8	3.4	3.2	3.5	3.5	3.6	3.5

By calculating the mean and ranges of the samples, find the action and warning limits for a quality control charts with $d = 2.06$. Sketch the chart, what is the tolerance level?

- 2 (a) A certain part has an exponential life distribution with a mean life of 1000 hours. What is the probability that such a part last at least 500 hours
- (b) A system consist of 6 components connected as shown. Find the overall reliability of the system if the reliabilities of A, B, C, D, E and F are 0.95, 0.85, 0.94, 0.99, 0.96 and 0.75 respectively



(c) Differentiate between Reliability and Validity

- 3 (a) State the six (6) properties of normal distribution.
- (b) X and Y are independent random variables which are normally distributed, the respective parameters being given in the table below:

	mean	variance
X	80	15
Y	90	10

Calculate the probability that $3X \leq 2Y + 80$

(c) Suppose that 6% of the items produced by a certain firm are defective. If 800 of such items are taken to be inspected. What is the probability that the number of defectives lies between 40 and 56?

4.

- (a) A continuous random variables has density function defined by
- $$f(x) = \begin{cases} Ky & 0 \leq y \leq 1 \\ Kx^{-5} & y > 1 \end{cases}$$
- Find K and standard deviation.

(b) A company wishes to purchase one of the five different machines. In an experimental designed to test whether there is a difference in the machine's performance each of the five experimental operators work on each of the machines. The number of units produced per machines are shown on the table below.

A	64	65	70	68	53
B	60	82	64	75	72
C	48	61	57	64	50
D	72	53	63	53	45
E	68	72	77	42	53

Test the hypothesis that there is no difference between the machines at 5% level of significance with $F_{95} = 1.78$

- (c) Which type of arrangement is used in this experimental designed?
5. (a) Find the expected number of girls on a committee of 3 selected at random from 3 girls and 4 boys.
- (b) A sample of 3 items is selected at random from a box containing 12 items of which 3 are defective. Find the expected number of defective item?
- (c) Team A has probability of $2/3$ if winning when ever the play. If A plays 4 games. Find the probability that A wins
- (i) exactly 2 games (ii) more than half of the game (iii) at least one game
6. (a) State the properties of Poisson Distribution
- (b) The average number of traffic accident on a certain highway is a 2 per week. Find the probability
- (i) no accidents on the 1st week period.
- (ii) At most 3 accidents on this highway during a 2 week period.
- (c) A coin is tossed 400 times. Find the probability that the number of heads will be
- (i) between 220 and 230 (ii) less than 210

$$\frac{\binom{2}{2} \binom{4}{3}}{\binom{7}{3}}$$

The Federal Polytechnic, Mubi

Department of Mathematics and Statistics, 2nd And 4th Semester Examinations, 2021/2022 Session

Departments: EE/ME/ABEET Course Code: MTH 313/412/413 Course Title: Statistical Methods

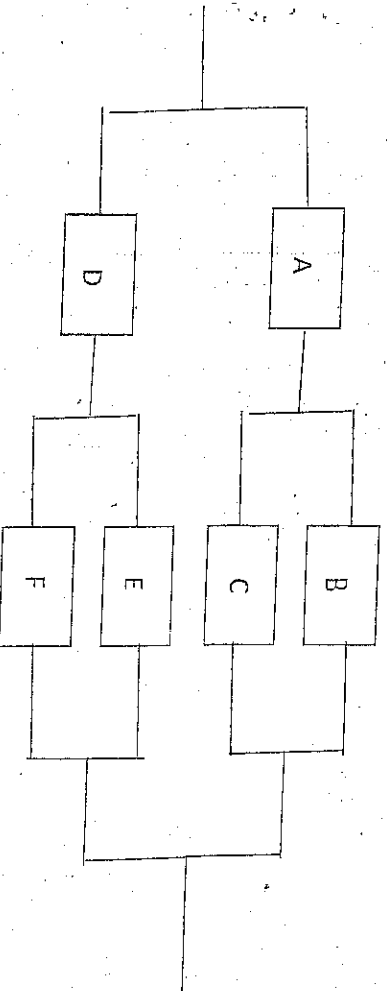
Instruction: Answer any Four (4) questions. Time allowed: 2hrs.

1

- (a) Explain the advantage of using quality control in the production of products
(b) Quality control is to be used for the samples in the table below. Calculate the quality control limits for the quality control chart with $d = 2.53$ for the tolerance limits

Sample No	1	2	3	4	5
	2.9	3.6	3.3	3.1	3.2
	3.2	3.6	3.6	3.0	2.9
	3.3	3.5	3.3	3.3	3.0
	3.7	3.8	3.0	3.9	3.2
	3.3	3.2	3.3	3.3	3.5
	3.3	3.7	3.1	3.3	3.2

2. (a) A system consist of 6 components connected as shown. Find the overall reliability of A, B, C, D, E and F are respectively 0.95, 0.75, 0.90, 0.95, 0.85 and 0.94



- (b) A system consists of 8 identical components in series. What must be the reliability of each component if the overall reliability is to be 0.95?

3. (a) Table below shows the yield per acre of rice grown in a particular soil treated with chemicals A, B, and C

(i) Find the mean yield for the different treatment
(ii) The grand mean for all the treatment
(iii) The total variation
(iv) The variation between the treatments
(v) The variation within the treatments

A	48	49	50
B	47	49	48
C	49	51	50

- (b) The Farmer wishes to test the effect of four different fertilizers A, B, C, and D on the yield of wheat. In order to eliminate sources of error due to variation in soil fertility. Determine whether there is difference between the fertilizer with $F_p = 4.76$

A18	C21	D25	B11
D22	B12	A15	C19
B15	A20	C23	D24

4. (a) A continuous random variables defined by

$$f(x) = \begin{cases} Kx & 0 \leq x \leq 2 \\ K(4-x)^2 & 2 \leq x \leq 4 \end{cases}$$

Determine the expected value of x.

- (b) The number of a success of an experiment is 0.6, the number of independent trials needed so that the expected value of successes is 9
- (c) Find the expected number of boys on a committee if 3 selected at random from 4 boys and 3 girls.
5. (a) State the properties of binomial distribution.
- (b) The probability that a team wins one match is 0.7, find the probability that they wins at most 6 matches out of 8
- (c) Suppose that 2% of the items made by a factory are defectives, find the probability that there are 3 defectives items in a sample of 200 items.
6. (a) State the properties of normal distribution.
- (b) Among 10,000 random digits, find the probability that the digit 3 appears at most 950 times
- (c) The scores of students are normally distributed with mean 60% and standard deviation of 5.6%.
- Find the probability that a given student scores
- (i) Less than 65%
- (ii) More than 70%
- (iii) Between 0% and 60%